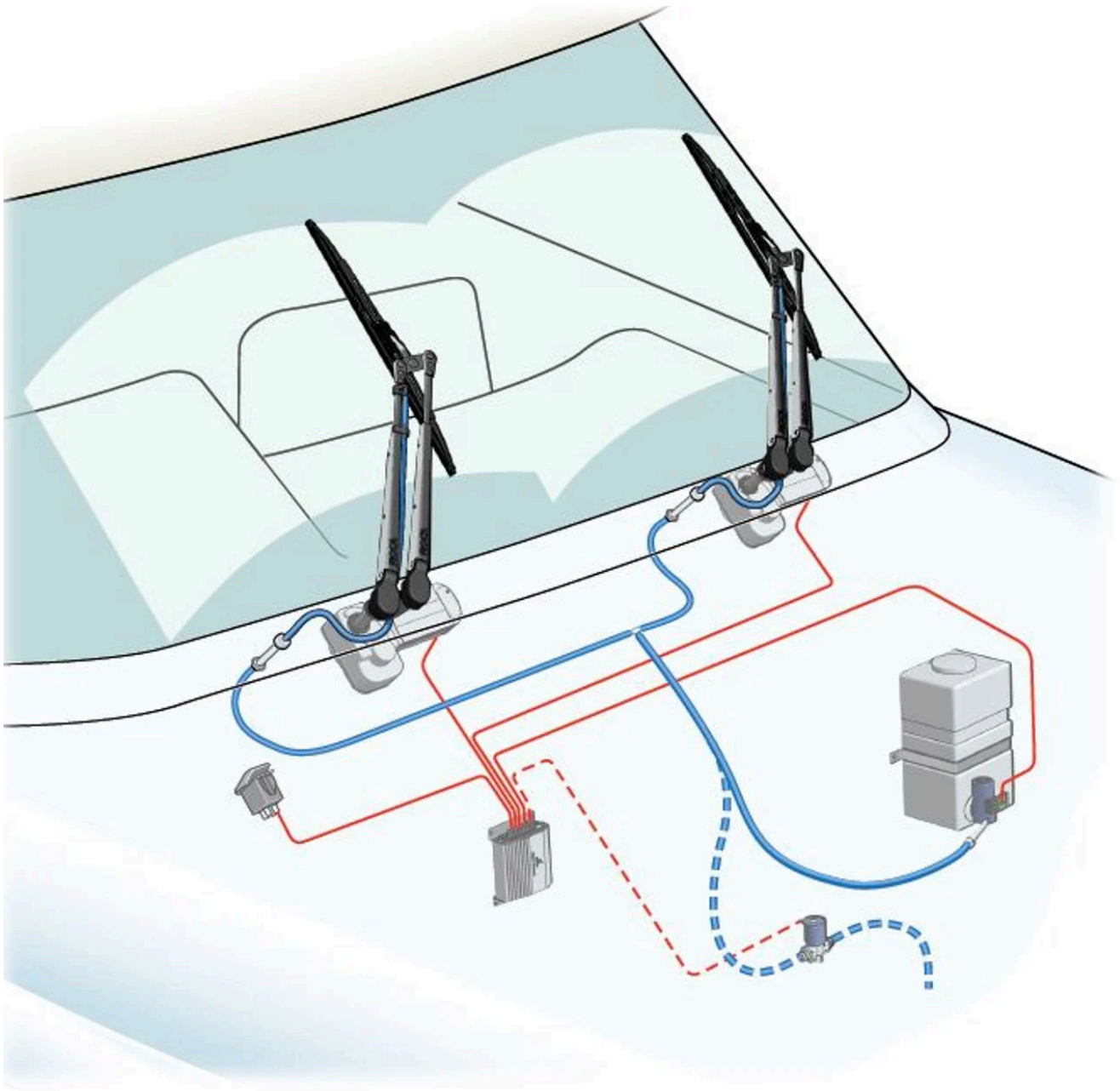


Wiper system Project description:



1. Project Objective

The objective is to develop a Simulink model that implements the control logic for a vehicle wiper system. The system must interpret driver inputs via a stalk switch (Off, Auto, Low, High), handle rain sensor error states, and generate the appropriate Pulse Width Modulation (PWM) command to drive the wiper motor.

2. System Architecture and Scheduling

The Simulink model must adhere to a specific root-level structure:

- **Main Subsystem:** The core control logic must be encapsulated within a subsystem named `RainSnsrWiprMot`.

- **Scheduling:** The control functionality must be executed every **10 milliseconds (10ms)**. This is managed by a function-call generator (`sample_time = 10e-3`) triggering the `Scheduler_10ms` port on the main subsystem.

3. Interface Definition (I/O)

3.1. Control Inputs

The system receives three primary inputs:

- `WiprMod` : Indicates the driver-selected wiper mode of operation.
 - 0: Off
 - 1: Aut (Automatic)
 - 2: LoSpd (Low Speed)
 - 3: HiSpd (High Speed)
- `RainSnsrErr` : A boolean flag indicating the health status of the rain sensor.
 - 0: Normal (no error)
 - 1: Error
- `WiprSpdReq` : An integer value representing the required speed level requested by the rain sensor during automatic mode.
 - Range: [0 1 2 3 4 5 6 7] that corresponds to [0 40% 45% 50% 55% 60% 65% 70%]

3.2. Control Outputs

The system must generate two outputs:

- `WiprMotPwmDutyCyc` : The PWM command signal sent to the wiper motor to control its speed.
- `WiprActv` : A status flag indicating if the wiper motor is currently running or activated.
 - 0: Stop (not activated)
 - 1: Running (activated)

4. Functional Requirements (Control Processing)

The control logic is determined primarily by the state of the `WiprMod` input.

4.1. Mode 0: Off

- The `WiprMotPwmDutyCyc` must be set to 0%.

4.2. Mode 2: Low Speed (LoSpd)

- The `WiprMotPwmDutyCyc` must be set to a constant 40%.

4.3. Mode 3: High Speed (HiSpd)

- The `WiprMotPwmDutyCyc` must be set to a constant 70%.

4.4. Mode 1: Automatic (Aut)

This mode involves conditional logic based on the rain sensor's health:

- **Error Handling:** If `RainSnsrErr` is true (1), the `WiprMotPwmDutyCyc` must be forced to zero, regardless of the speed request.
- **Normal Operation:** If `RainSnsrErr` is false (0), the system must map the `WiprSpdReq` input to a target PWM duty cycle using the following table:
 - Req 0: 0%
 - Req 1: 40%
 - Req 2: 45%
 - Req 3: 50%
 - Req 4: 55%
 - Req 5: 60%
 - Req 6: 65%
 - Req 7: 70%

5. Performance Requirements

- **PWM Smoothing:** During Automatic mode operation, the system is required to smooth the PWM command to avoid abrupt changes in motor speed when the `WiprSpdReq` changes. An appropriate rate limiter or filter should be implemented to achieve this transition.

6. Project Deliverables

- A complete Simulink model (.slx) implementing the architecture defined in Section 2.
- The internal control logic within the `RainSnsrWiprMot` subsystem satisfying all functional and performance requirements defined in Sections 4 and 5.
- Verification results demonstrating correct behavior for all modes, error conditions, and the required PWM smoothing in automatic mode.